

6/11/2026

Unit-1 Introduction to Big data and Hadoop.

Open source Framework to process large amt of data

BDA

BIG DATA ANALYTICS

"Dang cutting" & "Pike catering" - Father of Big data, 2005

• Analytics types:

- Analytics refers to systematic computational analysis of data.

- 4 types:

- 1) Descriptive analytics - What happened?
- 2) Diagnostic " - Why did it happen?
- 3) Predictive " - What will happen?
- 4) Prescriptive " - What should we do?

• Types of Digital Data:

1) structured data:

- Organized in rows & column (Ex: Relational database).
- standard format.
- Easy to store

Ex: Excel.

2) Semi-structured data:

- Doesn't follow strict schema but has tags/markers

(Ex: XML, JSON)

3) Unstructured data:

- No predefined format (Ex: text, img, video)
- Any type of data format.

• Big data:

Extremely large dataset cannot be processed by traditional tools.

characteristics (5 V's)

- 1) Volume - Huge amt of data
- 2) Velocity - speed of data generate & processing
- 3) variety - Diff type of data (structured / unstructured).
- 4) veracity - Data accuracy & reliability / Data may be messy
- 5) value - Extract meaningful insights.

Application:

- L Healthcare
- L Finance
- L Retail
- L social media
- L IOT

• Apache Hadoop ^{Software} Framework:

components:

HDFS (Hadoop distributed File system):
↓
store large file

MapReduce: Programming model for distributed data process

YARN (Yet Another Resource Negotiator): Resource management & job scheduling.

(RDBMS) Small data	Big data
* structured datatype - ex. excel * RDBMS * stored in Rows & column	* All type of data - Ex. vdo, text, audio * Hadoop.

Price water copper (PWC)

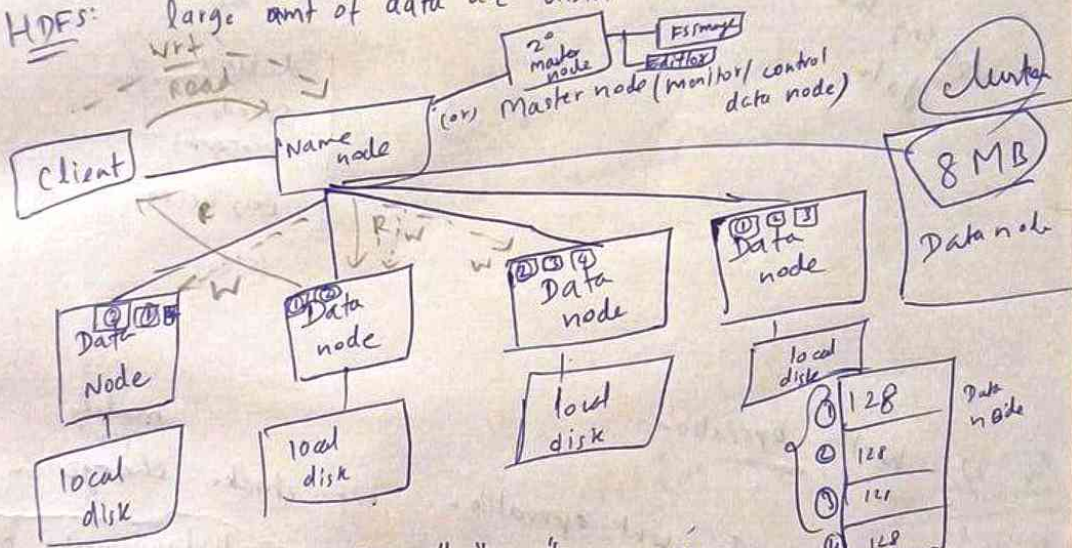
Price of book
Slavery
Copyright

- components of Hadoop:

- 1) HDFS - Hadoop distributed file system.
- 2) Mapping Reduce.
- 3) YARN - Yet Another resource negotiation.

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HDFS: large amt of data are distributed & stored.



Operations performed "read" & "write" operations. Ex:

limit of each data node - 128 MB

in oracle limit of each
data node - 64 KB

- cluster

cluster is adding extra block.

* Cluster are group of similar datatype where data nodes are placed in form of cluster

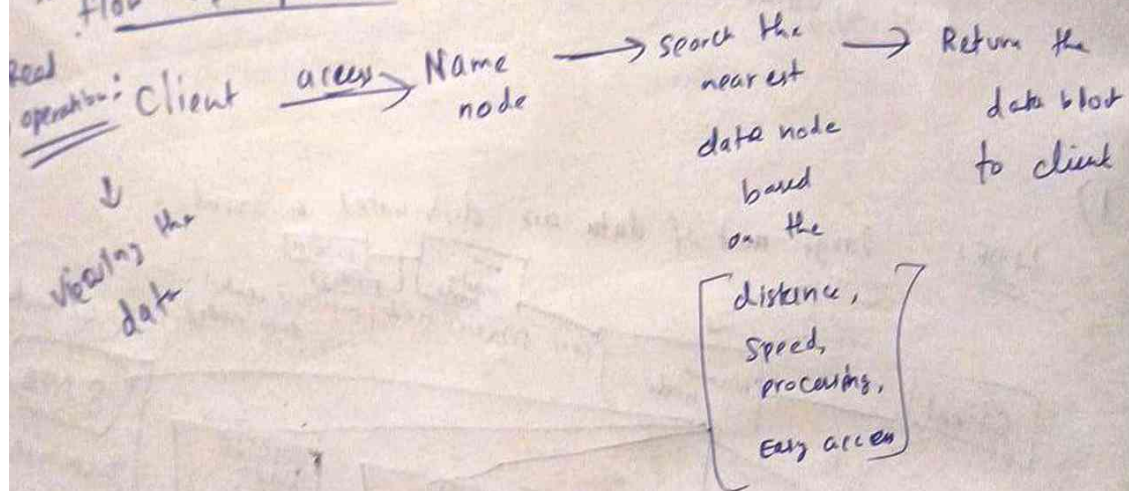
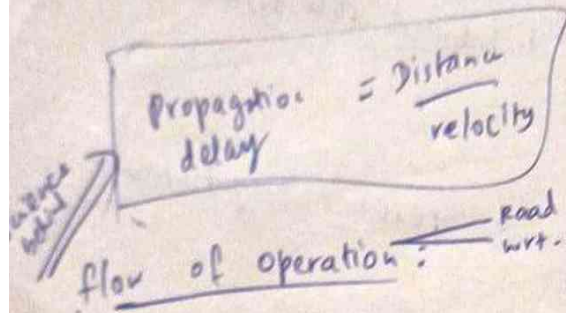
1mb of 1 block = 128
if distributed
4 block = 512
so remain
8 MB out
of 520
MB

8 MB stored
.. in cluster

1000's of TB = 1 Exa Byte

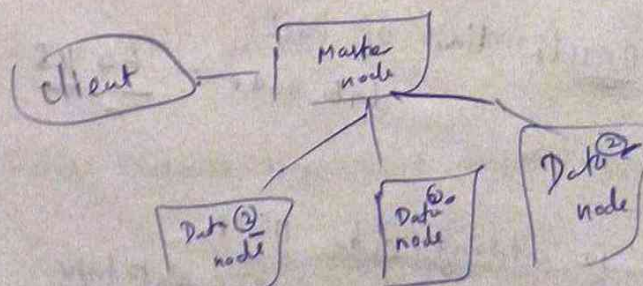
1000 GB = 1 Peta Byte

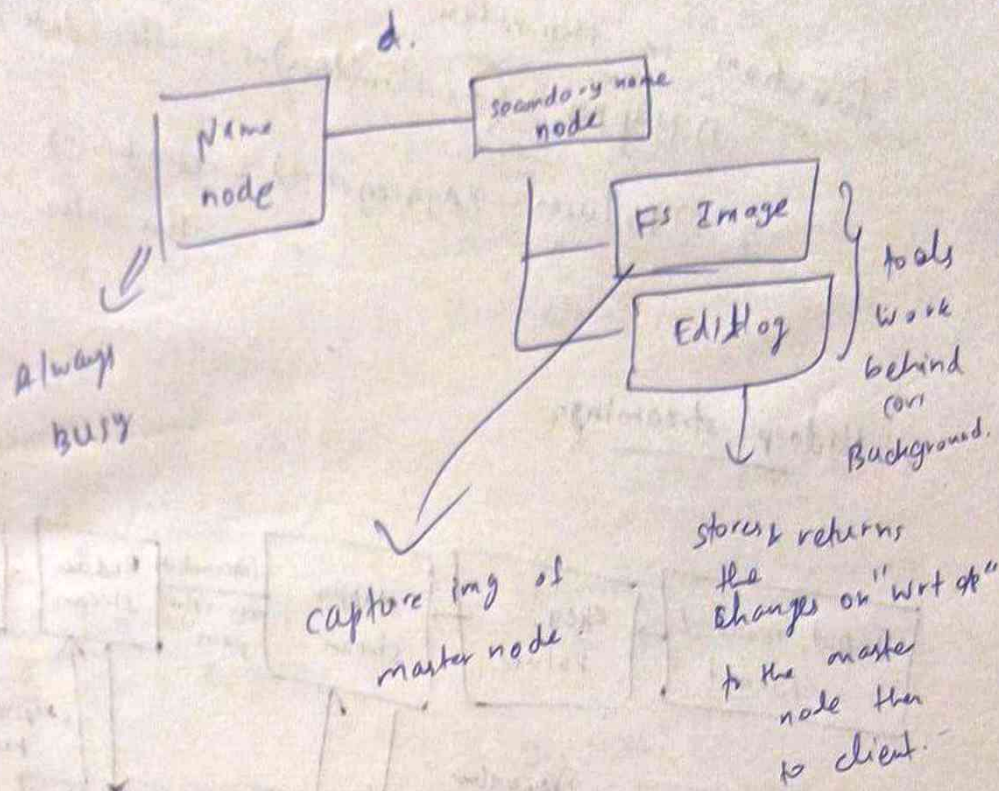
* Factor of HDFS is "Replication" copies of 3
of data



2) Write operation:

In wrt operation we make ^{modify (or)} changes on the data node where we parallelly make changes on other replica or copies stored in Data mode.





* FS Image:
It works on what kind of data are present at data nodes their locations & replica.

* EditLog:
Whatever changes you are performing during "Write operation" are stored in editlog.

2) Map reduce: (data processing)

steps:

Mapping/split - In format of $\Rightarrow$ Key: value
(key, value) $\Rightarrow$ 196: 352

Extract & organize - Asc/desc order

shuffle & sorting $\rightarrow$ 196: 352, 347
186: 357

Reduce $\Rightarrow$ Represent the "count" directly

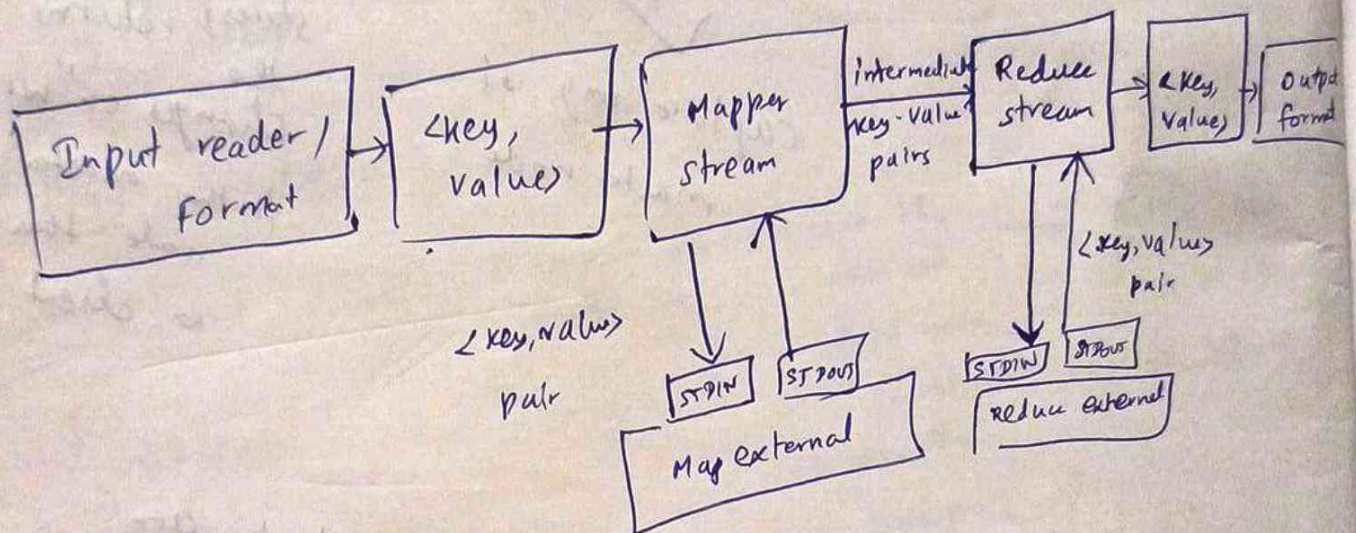
196: 2, 186: 1
repeated 2 times

key	val
user Id	movie Id
196	352
186	357
196	347

functions in Map reduce:

- 1) Mapper - (transformation) → transfer data from 1 form to another. i.e splitting
- 2) Reducer - (Aggregation) - combining the value

• Hadoop streaming:



In "Key: value" pair

always unique

can have multiple values.

DML command	DML command
create	Insert
alter	update
drop	delete
truncate	select